

City Hospital Evaluation

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Designation: COO City Hospital

Problem Summary

- No proper arrangements for patients
- Poor workload distribution for doctors
- Non availability of essential medications
- 15% re-test as a result of inconclusive results

Objectives

- To find the root cause of the crises City Hospital is currently facing by analyzing their data
- To gain clarity and insight from the data
- Provide recommendations based on the findings for efficient performance

Dataset. Overview

- 1 dataset used for the analysis
- Main columns used includes: Patient ID, Age group, Medical condition, Doctor, Date of admission, Test Results
- The data was cleaned to change the period in billing amount to comma

Key Metrics (KPIs)

- Total number of patients
- Total number of doctors
- Age group of patients
- Total duration (days) patient spent at the hospital

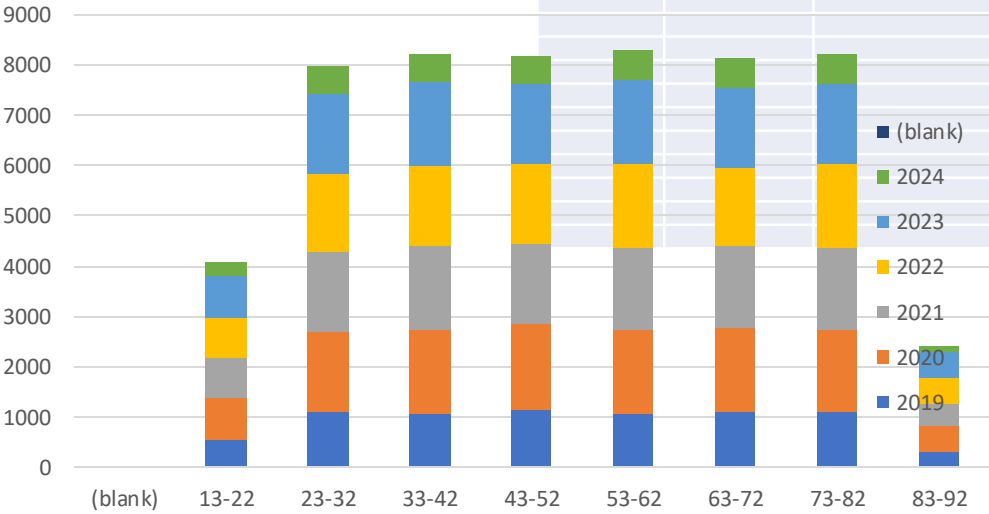
City Hospital Dashboard

CITY HOSPITAL DASHBOARD																			
1. PATIENTS OVERVIEW										2. ADMISSION TRENDS					3. SURVIVAL & DISCHARGE				
PATIENTS OVERVIEW										ADMISSION TRENDS					SURVIVAL & DISCHARGE				
Age Group	Gender	Admitted	Discharged	Transferred	Deceased	Readmitted	Refused	Waiting	Admitted	Discharged	Transferred	Deceased	Readmitted	Refused	Waiting	Admitted	Discharged	Transferred	Deceased
13-22	M	10	8	5	2	1	0	0	10	8	5	2	1	0	0	10	8	5	2
13-22	F	8	6	4	1	0	0	0	8	6	4	1	0	0	0	8	6	4	1
23-32	M	15	12	10	3	2	1	0	15	12	10	3	2	1	0	15	12	10	3
23-32	F	12	9	7	2	1	0	0	12	9	7	2	1	0	0	12	9	7	2
33-42	M	20	15	12	4	3	2	1	20	15	12	4	3	2	1	20	15	12	4
33-42	F	18	14	11	3	2	1	0	18	14	11	3	2	1	0	18	14	11	3
43-52	M	25	18	15	5	4	3	2	25	18	15	5	4	3	2	25	18	15	5
43-52	F	22	16	13	4	3	2	1	22	16	13	4	3	2	1	22	16	13	4
53-62	M	30	22	18	6	5	4	3	30	22	18	6	5	4	3	30	22	18	6
53-62	F	28	20	16	5	4	3	2	28	20	16	5	4	3	2	28	20	16	5
63-72	M	35	25	20	7	6	5	4	35	25	20	7	6	5	4	35	25	20	7
63-72	F	32	23	18	6	5	4	3	32	23	18	6	5	4	3	32	23	18	6
73-82	M	40	30	25	8	7	6	5	40	30	25	8	7	6	5	40	30	25	8
73-82	F	38	28	22	7	6	5	4	38	28	22	7	6	5	4	38	28	22	7
83-92	M	45	35	30	9	8	7	6	45	35	30	9	8	7	6	45	35	30	9
83-92	F	42	32	28	8	7	6	5	42	32	28	8	7	6	5	42	32	28	8
93-102	M	50	40	35	10	9	8	7	50	40	35	10	9	8	7	50	40	35	10
93-102	F	48	38	32	9	8	7	6	48	38	32	9	8	7	6	48	38	32	9
103-112	M	55	45	40	11	10	9	8	55	45	40	11	10	9	8	55	45	40	11
103-112	F	52	42	38	10	9	8	7	52	42	38	10	9	8	7	52	42	38	10
113-122	M	60	50	45	12	11	10	9	60	50	45	12	11	10	9	60	50	45	12
113-122	F	58	48	42	11	10	9	8	58	48	42	11	10	9	8	58	48	42	11
123-132	M	65	55	50	13	12	11	10	65	55	50	13	12	11	10	65	55	50	13
123-132	F	62	52	48	12	11	10	9	62	52	48	12	11	10	9	62	52	48	12
133-142	M	70	60	55	14	13	12	11	70	60	55	14	13	12	11	70	60	55	14
133-142	F	68	58	52	13	12	11	10	68	58	52	13	12	11	10	68	58	52	13
143-152	M	75	65	60	15	14	13	12	75	65	60	15	14	13	12	75	65	60	15
143-152	F	72	62	58	14	13	12	11	72	62	58	14	13	12	11	72	62	58	14
153-162	M	80	70	65	16	15	14	13	80	70	65	16	15	14	13	80	70	65	16
153-162	F	78	68	62	15	14	13	12	78	68	62	15	14	13	12	78	68	62	15
163-172	M	85	75	70	17	16	15	14	85	75	70	17	16	15	14	85	75	70	17
163-172	F	82	72	68	16	15	14	13	82	72	68	16	15	14	13	82	72	68	16
173-182	M	90	80	75	18	17	16	15	90	80	75	18	17	16	15	90	80	75	18
173-182	F	88	78	72	17	16	15	14	88	78	72	17	16	15	14	88	78	72	17
183-192	M	95	85	80	19	18	17	16	95	85	80	19	18	17	16	95	85	80	19
183-192	F	92	82	78	18	17	16	15	92	82	78	18	17	16	15	92	82	78	18
193-202	M	100	90	85	20	19	18	17	100	90	85	20	19	18	17	100	90	85	20
193-202	F	98	88	82	19	18	17	16	98	88	82	19	18	17	16	98	88	82	19
203-212	M	105	95	90	21	20	19	18	105	95	90	21	20	19	18	105	95	90	21
203-212	F	102	92	88	20	19	18	17	102	92	88	20	19	18	17	102	92	88	20
213-222	M	110	100	95	22	21	20	19	110	100	95	22	21	20	19	110	100	95	22
213-222	F	108	98	92	21	20	19	18	108	98	92	21	20	19	18	108	98	92	21
223-232	M	115	105	100	23	22	21	20	115	105	100	23	22	21	20	115	105	100	23
223-232	F	112	102	98	22	21	20	19	112	102	98	22	21	20	19	112	102	98	22
233-242	M	120	110	105	24	23	22	21	120	110	105	24	23	22	21	120	110	105	24
233-242	F	118	108	102	23	22	21	20	118	108	102	23	22	21	20	118	108	102	23
243-252	M	125	115	110	25	24	23	22	125	115	110	25	24	23	22	125	115	110	25
243-252	F	122	112	108	24	23	22	21	122	112	108	24	23	22	21	122	112	108	24
253-262	M	130	120	115	26	25	24	23	130	120	115	26	25	24	23	130	120	115	26
253-262	F	128	118	112	25	24	23	22	128	118	112	25	24	23	22	128	118	112	25
263-272	M	135	125	120	27	26	25	24	135	125	120	27	26	25	24	135	125	120	27
263-272	F	132	122	118	26	25	24	23	132	122	118	26	25	24	23	132	122	118	26
273-282	M	140	130	125	28	27	26	25	140	130	125	28	27	26	25	140	130	125	28
273-282	F	138	128	122	27	26	25	24	138	128	122	27	26	25	24	138	128	122	27
283-292	M	145	135	130	29	28	27	26	145	135	130	29	28	27	26	145	135	130	29
283-292	F	142	132	128	28	27	26	25	142	132	128	28	27	26	25	142	132	128	28
293-302	M	150	140	135	30	29	28	27	150	140	135	30	29	28	27	150	140	135	30
293-302	F	148	138	132	29	28	27	26	148	138	132	29	28	27	26	148	138	132	29
303-312	M	155	145	140	31	30	29	28	155	145	140	31	30	29	28	155	145	140	31
303-312	F	152	142	138	30	29	28	27	152	142	138	30	29	28	27	152	142	138	30
313-322	M	160	150	145	32	31	30	29	160	150	145	32	31	30	29	160	150	145	32
313-322	F	158	148	142	31	30	29	28	158	148	142	31	30	29	28	158	148	142	31
323-332	M	165	155	150	33	32	31	30	165	155	150	33	32	31	30	165	155	150	33
323-332	F	162	152	148	32	31	30	29	162	152	148	32	31	30	29	162	152	148	32
333-342	M	170	160	155	34	33	32	31	170	160	155	34	33	32	31	170	160	155	34
333-342	F	168	158	152	33	32	31	30	168	158	152	33	32	31	30	168	158	152	33
343-352	M	175	165	160	35	34	33	32	175	165	160	35	34	33	32	175	165	160	35
343-352	F	172	162	158	34	33	32	31	172	162	158	34	33	32	31	172	162	158	34
353-362	M	180	170	165	36	35	34	33	180	170	165	36	35	34	33	180	170	165	36
353-362	F	178	168	162	35	34	33	32	178	168	162	35	34	33	32	178	168	162	35
363-372	M	185	175	170	37	36	35	34	185	175	170	37	36	35	34	185	175	170	37
363-372	F	182	172	168	36	35	34	33	182	172	168	36	35	34	33	182	172	168	36
373-382	M	190	180	175	38	37	36	35	190	180	175	38	37	36	35	190	180	175	38
373-382	F	188	178	172	37	36	35	34	188	178	172	37	36	35	34	188	178	172	37
383-392	M	195	185	180	39	38	37	36	195	185	180	39	38	37	36	195	185	180	39
383-392	F	192	182	178	38	37	36	35	192	182	178	38	37	36	35	192	182	178	38
393-402	M	200	190	185	40	39	38	37	200	190	185	40	39	38	37	200	190	185	40
393-402	F	198	188	182	39	38	37	36	198	188	182	39	38	37	36	198	188	182	39
403-412	M	205	195	190	41	40	39	38	205	195	190	41	40	39	38	205	195	190	41
403-412	F	202	192	188	40	39	38	37	202	192	188	40	39	38	37	202	192	18	

Dashboard Insight

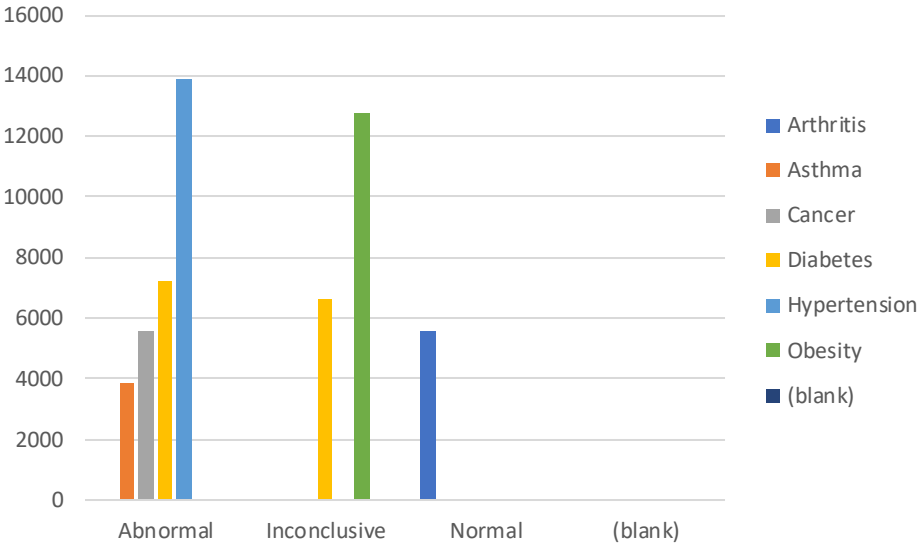
1. PATIENTS DEMOGRAPHICS									
KPIs									
Total patients									
Age distribution									
Count of Patient ID		Column Labels							
Row Labels		2019	2020	2021	2022	2023	2024(blank)		Grand Total
(blank)									
13-22		558	821	776	822	813	292		4082
23-32		1091	1604	1578	1545	1610	537		7965
33-42		1069	1662	1653	1610	1651	562		8207
43-52		1123	1729	1580	1598	1603	561		8194
53-62		1067	1656	1628	1678	1670	582		8281
63-72		1084	1684	1634	1567	1580	604		8153
73-82		1092	1629	1627	1670	1596	584		8198
83-92		303	500	455	527	503	132		2420
Grand Total		7387	11285	10931	11017	11026	3854		55500

Age range of Patients Admitted



Dashboard Insight

Count of Patient ID	Column Labels							
Row Labels	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity	(blank)	Grand Total
Abnormal		3885	5550	7215	13875			30525
Inconclusive				6660		12765		19425
Normal	5550							5550
(blank)								
Grand Total	5550	3885	5550	13875	13875	12765		55500
Count of Patient ID	Column Labels							
Row Labels	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity	(blank)	Grand Total
Abnormal		3885	5550	15369.2	13875			49395
Inconclusive				17205		12765		56055
Normal	5550			19040.8				62715
(blank)				20876.5				69375
Grand Total	5550	3885	5550	22712.3	13875	12765		76035
Count of Patient ID	Column Labels							
Row Labels	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity	(blank)	Grand Total
Abnormal		3885	5550	24548.1	13875			82695
Inconclusive				26383.8		12765		89355
Normal	5550			28219.6				96015
(blank)				30055.4				102675
Grand Total	5550	3885	5550	31891.2	13875	12765		109335
Count of Patient ID	Column Labels							
Row Labels	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity	(blank)	Grand Total
Abnormal		3885	5550	33726.9	13875			115995
Inconclusive				35562.7		12765		122655



Recommendations

- Most Patients admitted are within the age range of 23 to 82 years. Proper and adequate preparation has to be made for them.
- Use the medication metrics used by patients to ensure that the store is not out of essential drugs
- Average duration (in days) for patients can be used to distribute patient attendance workload for doctors
- To be able to get busiest hour at the hospital, time when patients are admitted have to be included in the record book.
- Emergency spikes expected at the first week of the year

Conclusion

- Thank you board for this opportunity.
- With the dashboard, we have better insight on the causes of crises the hospital is going through and how to fix it.